

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.TECH.

Branch : MECHANICAL ENGINEERING

Semester : III

Subject Code & Name: BTMEC302_Y19 MATERIALS SCIENCE AND METALLURGY

Max Marks: 60

Date: 07/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	Line Defects are also known as		1				
	<table border="1"> <tr> <td>a. Stacking Fault</td> <td>b. Point Defect</td> <td>c. Volume Defect.</td> <td>d. Dislocations</td> </tr> </table>	a. Stacking Fault	b. Point Defect	c. Volume Defect.	d. Dislocations		
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2	Rise in stress during plastic deformation is called		1				
	<table border="1"> <tr> <td>a. Toughness</td> <td>b. Hardness</td> <td>c. work Hardening</td> <td>d. Stress Relieving</td> </tr> </table>	a. Toughness	b. Hardness	c. work Hardening	d. Stress Relieving		
a. Toughness	b. Hardness	c. work Hardening	d. Stress Relieving				
3	Hardness test may give the information about		1				
	<table border="1"> <tr> <td>a. wear Resistance</td> <td>b. Machinability</td> <td>c. Effectiveness of the heat treatment</td> <td>d. All of these</td> </tr> </table>	a. wear Resistance	b. Machinability	c. Effectiveness of the heat treatment	d. All of these		
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4	The failure that happen due to cyclic loading is called....		1				
	<table border="1"> <tr> <td>a. Creep Failure</td> <td>b. Fatigue Failure</td> <td>c. Shears Failure</td> <td>d. Tensile Failure</td> </tr> </table>	a. Creep Failure	b. Fatigue Failure	c. Shears Failure	d. Tensile Failure		
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5	Phase diagram is obtained by		1				
	<table border="1"> <tr> <td>a. Very Fast Cooling</td> <td>b. Very Fast Heating</td> <td>c. Very Slow Cooling</td> <td>d. Initially Cooling and then Heating</td> </tr> </table>	a. Very Fast Cooling	b. Very Fast Heating	c. Very Slow Cooling	d. Initially Cooling and then Heating		
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6	According to the Hume Ruther which factor does not govern solid solubility?		1				
	<table border="1"> <tr> <td>a. Crystal Structure</td> <td>b. Relative Size</td> <td>c. Relative Density</td> <td>d. Relative Valency</td> </tr> </table>	a. Crystal Structure	b. Relative Size	c. Relative Density	d. Relative Valency		
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7	Which of the following constituents of steels is softest and least strong.....		1				
	<table border="1"> <tr> <td>a. Austenite</td> <td>b. Pearlite</td> <td>c. Cementite</td> <td>d. Ferrite</td> </tr> </table>	a. Austenite	b. Pearlite	c. Cementite	d. Ferrite		
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8	Which type of test is mostly used for detecting surface defects in crane hooks		1				
	<table border="1"> <tr> <td>a. Magnaflux</td> <td>b. Ultrasonic</td> <td>c. Radiography</td> <td>d. None of these</td> </tr> </table>	a. Magnaflux	b. Ultrasonic	c. Radiography	d. None of these		
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9	Water proof Emery papers are used for					1
	a. Dry Polishing	b. Wet Polishing	c. Fine Polishing	d. Electrolytic Polishing		
10	Powder Metallurgy is most suitable method for ----- Hardened Materials.					1
	a. Dispersion	b. Age	c. Strain	d. Precipitation		
11	Quenching is not required in -----					1
	a. Carburizing	b. Nitriding	c. Carbonitriding	d. Hardening		
12	Which one may be used as a material for Transducer in Ultrasonic Test?					1
	a. Steel	b. Qurtz	c. Polymer	d. None of these		
Q. 2 Solve the following.						12
A)	Derive the Expressions for planar atomic on (100), (110) and (111) planes of both BCC and FCC crystal structures.					6
B)	Discuss Induction hardening process w.r.t. principle of working process details advantages and limitations.					6
Q.3 Solve the following.						12
A)	Explain dispersion strengthening w.r.t. basic mechanisms, critical factors advantages and commercial examples.					6
B)	Explain the principle of working of optical metallurgical Microscope. Compare it with electron microscope.					6
Q. 4 Solve Any Two of the following.						12
A)	Describe the procedure of specimen preparation of microscopy.					6
B)	What is the importance of T-T-T diagrams? Explain the procedure to determine these diagrams with the help of schematic diagrams					6
C)	Explain the principle of magnetic particle testing (magna flux) and what are the methods of magnetization? state advantages and disadvantages of demagnetization					6
Q.5 Solve Any Two of the following.						12
A)	Define heat treatment and give its objective. Give the names of two different heat treatment processes along with the major objective of each.					6
B)	Draw Fe-Fe ₃ equilibrium diagram and explain various phases and critical temperatures present in it.					6
C)	Discus Sulphur print test w.r.t. purposes, significance, procedure, and chemical reactions.					6

Q. 6	Solve Any Two of the following.		12
A)	Describe ultrasonic inspection technique w.r.t principle of working types, and applications		6
B)	Discuss Vickers Hardness tests w.r.t. principle of working, indenter details, formula, advantages and limitations.		6
C)	Explain Slip Mechanism of Plastic Deformation w.r.t. conceptual meaning its occurrence due to the movement of edge and screw dislocations and comparison with twinning.		6
*** End ***			

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech. Branch : Mechanical Engineering/Mechanical Engineering (Sandwich)

Subject Code & Name: BTMES304; Materials Science and Metallurgy Semester : III

Max Marks: 60

Date: 12/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
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4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
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		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	There are fourteen atoms in a unit cell of	CO1	1
	a. BCC b. FCC c. HCP d. none of these		
2	The elastic stress strain behavior of rubber is	CO1	1
	a. Linear b. Non-linear c. Plastic d. No fixed relationship		
3	Substitution of a foreign atom in the site of parent atom in the crystal is a	CO1	1
	a. Vacancy defect b. Substitution impurity c. Volume imperfection d. Line imperfections		
4	Gibbs phase rule is, when no chemical reaction occurs	CO3	1
	a. $F - C + P = 2$ b. $F + C - P = 2$ c. $F + C + P = 2$ d. $F - C - P = 2$		
5	The percentage of carbon in cast iron varies from...	CO2	1
	a. 0.1 to 0.5 b. 0.5 to 1 c. 0.5 to 1 d. 2 to 4.5		
6	Steel containing 0.8 to 1.5% carbon, is known as...	CO2	1
	a. mild steel b. dead mild steel c. medium carbon steel d. high carbon steel		
7	A given component cracked after heat treatment. What can be the possible reason?	CO4	1
	a. It was heated b. It was not c. It was d. It was slowly		

	for long time	properly cleaned before heating	suddenly cooled in brine	cooled in air		
8	The process of producing a component with tough and ductile core and a hard outer surface is known as...				CO4	1
	a. Hardening	b. Case Hardening	c. Tempering	d. Annealing		
9	The study of metallographic includes.....				CO5	1
	a. alloy constituents	b. failure analysis	c. metal structure	d. all of the above		
10	In order to observe the grain size of steel samples under microscope, the magnification should be the order of....				CO5	1
	a. 2	b. 20	c. 1500	d. 100		
11	Dye penetrant method is generally used to locate...				CO6	1
	a. core defects	b. surface defects	c. superficial defects	d. temporary defects		
12	During radiography test, which region absorbs less radiation and transmits more?				CO6	1
	a. Low and high density regions absorb and transmit same amount of radiation	b. High density region	c. Low density region	d. None of the above		
Q.2	Solve the following.					12
A)	Explain stress strain curves for different materials with figures.				CO2	6
B)	Explain Imperfections in crystals.				CO1	6
Q.3	Solve the following.					12
A)	Draw neat and complete Iron-iron carbide equilibrium diagram, showing phases, temperatures.				CO3	6
B)	Describe classifications, and applications of different steels.				CO3	6

Q. 4	Solve Any Two of the following.		12
A)	Explain tensile testing method with stress-strain curve and necessary formulas.	CO2	6
B)	Describe the normalizing heat treatment process with neat figure.	CO4	6
C)	Describe the nitriding heat treatment process with neat figure.	CO4	6
Q.5	Solve Any Two of the following.		12
A)	Explain in brief how to prepare specimen for metallography testing.	CO5	6
B)	Describe Spark test with figures.	CO5	6
C)	Draw CCT diagram for steel and write its importance.	CO3	6
Q. 6	Solve Any Two of the following.		12
A)	Differentiate between destructive testing and non-destructive testing.	CO6	6
B)	Explain ultrasonic inspection with neat figure.	CO6	6
C)	Explain any one Strengthening method for metals.	CO6	6
*** End ***			